

“Water, milk and minimally processed foods should remain the foundation of the diet, rather than routinely relying on highly or artificially sweetened packaged products”

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processed foods should remain the foundation of the diet, rather than routinely relying on highly sweetened or artificially sweetened packaged products,” says Dr Eileen Canday, head, department of nutrition & dietetics, Sir H.N. Reliance Foundation Hospital, Mumbai.

The bigger surprise may be that some of the most closely studied ingredients are not sweeteners at all. Carboxymethylcellulose, or CMC (E466), is a cellulose-derived emulsifier and stabiliser that helps keep ingredients from separating and gives ice creams, sauces and other processed foods their smooth texture. Human studies have raised questions about its effects on the gut. In a small, controlled trial, people consuming CMC experienced changes in the composition and diversity of their gut bacteria, along with changes in the gut’s metabolic by-products. “Research indicates that some emulsifiers and stabilisers may affect the microbial composition of the gut, the gut barrier and inflammatory processes,” says Dr Samiksha Kalra, dietician and lactation consultant at Madhukar Rainbow Children’s Hospital, Delhi. Nor does approval mean a food containing an additive is automatically healthy. “It is important to keep in mind that the fact that an additive has been approved means it has passed tests for safety; this does not mean that all foods containing it are healthy,” says Dr Deepali Sharma, clinical nutritionist at the CK Birla Hospital, Delhi. Also, limits apply.

The same is true of the innocuous-sounding ‘modified starch’. It starts as ordinary starch but is altered physically, enzymatically or chemically so that soups stay thick, fillings do not separate and frozen foods retain their texture after reheating. Modified starches featured prominently in one of the additive mixtures associated with a higher incidence of type 2 diabetes in a 2025 *PLOS Medicine* study. But the finding was about the mixture as a whole—not modified starch alone—which included pectin, guar gum, carrageenan, polyphosphates, potassium sorbate, curcumin and xanthan gum.

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Then there are ingredients consumers rarely think about at all, such as phosphate additives. Sodium phosphates, polyphosphates and phosphoric acid are used in processed foods to control acidity, retain moisture and maintain texture. The concern is particularly relevant for people with impaired kidney function, because the kidneys play a central role in regulating phosphorus levels. Research has found that reducing phosphate additives can lower hormones involved in phosphorus regulation, with effects that may be particularly relevant in people with chronic kidney disease.

THE PROTEIN TRAP

At the opposite end of the modern health aisle are protein isolates and concentrates—whey, soy, pea or wheat protein separated from the rest of the food and concentrated into a powder. They are not inherently harmful and can be useful when extra protein is genuinely needed. The problem is what else comes with them. A product may be labelled ‘protein’, ‘multigrain’, ‘baked’ or ‘no added sugar’ and still contain refined ingredients, sodium, saturated fat, sweeteners or other additives. “I have seen frequent consumption of packaged foods contributes to childhood obesity, insulin resistance, early signs of fatty liver disease, constipation and other digestive problems,” says Dr Kanchankumar Bhagyawant, consultant paediatrician and neonatologist at Ruby Hall Clinic, Pune.

Perhaps the most important development, however, is that researchers are beginning to look beyond individual ingredients and study the mixtures people actually consume. “Real people do not eat one additive in a laboratory; they eat mixtures. So, we don’t know if this emulsifier caused this condition, but we do know that higher consumption of UPF is associated with adverse health outcomes, even when the food is marketed as healthy,” says Dr Wasir. Recent research is increasingly trying to untangle how these combinations—and the foods that carry them—may affect health. That may be the real warning hidden inside the modern packet. ■